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Last time. $G'=G^{(1)}=G^{der}=[G,G]:=\langle\{[x,y]=x^{-1}y^{-1}xy \mid x,y\in G\}\rangle \lhd G$

G/G' abelian

对$M\lhd G$, G/M交换$\Leftrightarrow xM\cdot yM=yM\cdot xM, \forall x,y\in G \Leftrightarrow [x,y]\in M, \forall x,y\in G \Leftrightarrow G'<M$.

G/G' is the "maximal abelian quotient" of G 则G/M交换$\Leftrightarrow G'<M$.

LEM For A an abelian group and $\phi: G\to A$, we have $G'<\ker\phi$. ($\phi(x^{-1}y^{-1}xy)=1$)

Then ϕ factors as $G\xrightarrow{\pi} G/G'\xrightarrow{\bar{\phi}} A$.

- (universal property)

$$\begin{array}{ccc} G & \xrightarrow{\phi} & A \\ {\scriptstyle\pi}\downarrow & \nearrow_{\exists!\,\bar{\phi}} & \\ G/G' & & \end{array}$$

- get bijection $\mathrm{Hom}(G,A)\simeq\mathrm{Hom}(G/G',A)$, $\phi\mapsto\bar{\phi}$

eg. $G=D_{2n}=\langle r,s \mid r^n=s^2=1, srs=r^{-1}\rangle$

G' contains $s^{-1}r^{-1}sr=r\cdot r=r^2$.

- if n is odd $\quad\langle r^2\rangle=\langle r\rangle<G'$

define $\phi: G\to\{\pm1\}$ ← abelian homomorphism

$r\mapsto +1$

$s\mapsto -1$ "orientation".

LEM $\Rightarrow G'<\ker\phi=\langle r\rangle \Rightarrow G'=\langle r\rangle$.

- if n is even $\quad\langle r^2\rangle<G'$

define $\phi: G\to\{\pm1\}\times\{\pm1\}$ ← abelian hom.

$r\mapsto(1,-1)$

$s\mapsto(+1,1)$

(Check $\phi(r)^n=\phi(s)^2=1$, $\phi(s)\phi(r)\phi(s)=\phi(r)^{-1}$)

LEM $\Rightarrow G'<\ker\phi=\langle r^2\rangle\Rightarrow G'=\langle r^2\rangle$ $\quad G/G'\simeq\{\pm1\}\times\{\pm1\}$.

Solvable groups

Recall. A group G is called solvable if $\exists\ \{e\}=G_0\lhd G_1\lhd\cdots\lhd G_k=G$, st. G_i/G_{i-1} is abelian, $\forall 1\le i\le k$.

$\Updownarrow$

$G_i'<G_{i-1}$

Testing solvability by derived series.

DEF. For a group G, define inductively a sequence of subgroup

$G^{(0)}=G$, $G^{(1)}=[G,G]$, $\cdots$, $G^{(i+1)}=[G^{(i)},G^{(i)}]$, $\forall i$.

$\rightsquigarrow G=G^{(0)}\triangleright G^{(1)}\triangleright\cdots$ called the derived series of G (导出序列)

PROP A group G is solvable $\Leftrightarrow G^{(n)}=\{1\}$ for some finite n.

Pf "$\Leftarrow$": $\{1\}=G^{(n)}\triangleleft G^{(n-1)}\triangleleft\cdots\triangleleft G^{(0)}=G$ $\quad G^{(i)}/G^{(i+1)}$ is abelian, $\forall i$. $\Rightarrow G$ solvable.

"$\Rightarrow$": G solvable $\Rightarrow \exists\{1\}=H_0\triangleleft H_1\triangleleft\cdots\triangleleft H_k=G$, st. H_i/H_{i-1} is abelian, $\forall i$.

$\Rightarrow [H_i,H_i]<\ker(H_i\to H_i/H_{i-1})=H_{i-1}$ (abelian) 比较序列的"下降速度"

Now $G^{(1)}=[G,G]<H_{k-1}$

$G^{(2)}=[G^{(1)},G^{(1)}]<[H_{k-1},H_{k-1}]<H_{k-2}$

$\cdots\quad G^{(k)}<H_0=\{1\}$. □

REM The derived series is the "fastest decreasing" series whose successive quotients are abelian.

The smallest $n\in\mathbb{N}$ for which $G^{(n)}=\{1\}$ is called the solvable length of G.

eg. $G=\left\{\begin{pmatrix}*&*&*\\0&*&*\\0&0&*\end{pmatrix}\right\}\triangleright G^{(1)}=\left\{\begin{pmatrix}1&*&*\\0&1&*\\0&0&1\end{pmatrix}\right\}\triangleright G^{(2)}=\left\{\begin{pmatrix}1&0&*\\0&1&0\\0&0&1\end{pmatrix}\right\}\triangleright G^{(3)}=\{1\}$

LEM All $G^{(i)}$ are normal/characteristic subgps of G.

Pf $\forall\phi: G\xrightarrow{\sim}G$ automorphism

$\phi(G^{(1)})=\langle\{\phi(x^{-1}y^{-1}xy)\mid x,y\in G\}\rangle=\langle\{\phi(x)^{-1}\phi(y)^{-1}\phi(x)\phi(y)\mid x,y\in G\}\rangle=G^{(1)}$

inductively $\phi(G^{(i)})=\phi([G^{(i-1)},G^{(i-1)}])=[\phi(G^{(i-1)}),\phi(G^{(i-1)})]=[G^{(i-1)},G^{(i-1)}]=G^{(i)}$ □

Easy properties

1). if $N\triangleleft G$ and both N, G/N are solvable, then G is solvable.

2). $H<G\Rightarrow H^{(i)}<G^{(i)}$ so G is solvable $\Rightarrow H$ is solvable

3). $\phi: G\twoheadrightarrow K$ surjective homomorphism $\Rightarrow\phi(G^{(i)})=K^{(i)}$ so G is solvable $\Rightarrow K$ is solvable.

THM (Feit-Thompson 1963) All finite groups of odd order are solvable.

Nilpotent groups

DEF A group G is nilpotent if it admits a sequence of subgroups

$\{1\}=H_0\triangleleft H_1\triangleleft\cdots\triangleleft H_k=G$, st. $[G,H_i]<H_{i-1}$, $\forall 1\le i\le k$.

This is called a central series of G. Here $[G,H]=\langle\{g^{-1}h^{-1}gh \mid g\in G, h\in H\}\rangle$.

($[G,H_i]<H_{i-1}<H_i \Rightarrow$ all H_i are normal subgps of G

$[G,H_i]<H_{i-1} \Leftrightarrow [G/H_{i-1}, H_i/H_{i-1}]=\{1\} \Leftrightarrow H_i/H_{i-1}<Z(G/H_{i-1})$ ← abelian, ← abelian)

in particular, G nilpotent $\Rightarrow G$ solvable.

{cyclic group}⊂{abelian groups}⊂{nilpotent groups}⊂{solvable groups}

! nonabelian simple groups are NOT solvable. (Feit-Thompson). → 事实上不需要. 对非交换单群G有$\{1\}\neq G'\lhd G \Rightarrow G'=G$. 一直求导到不了$\{1\}$, 故不可解.

Testing nilpotency by lower/upper central series.

DEF For a group G define inductively a sequence of subgps.

$G^0=G$, $G'=[G,G]$, $\cdots$, $G^{i+1}=[G,G^i]$, $\forall i$

$\rightsquigarrow G=G^0\rhd G'\rhd\cdots$ called the lower central series of G

same as arbitrary central series. all G^i are normal subgps of G.

$G^i/G^{i+1}<Z(G/G^{i+1})$ and $G^i>G^{(i)}$, $\forall i$

"Dually"

DEF For a group G, define inductively $Z_0=\{1\}$, $Z_1=Z(G)$

Consider $\pi_1: G\to G/Z_1\rhd Z(G/Z_1)$

Put $Z_2:=\pi_1^{-1}(Z(G/Z_1))\lhd G$ $\quad(Z_2/Z_1=Z(G/Z_1))$

inductively $\pi_i: G\to G/Z_i\rhd Z(G/Z_i)$

Put $Z_{i+1}:=\pi_i^{-1}(Z(G/Z_i))$ $\quad(Z_{i+1}/Z_i=Z(G/Z_i))$

$\rightsquigarrow \{1\}=Z_0\lhd Z_1\lhd\cdots$ called the upper central series of G.

PROP A group G is nilpotent $\Leftrightarrow G^n=\{1\}$ for some finite n $\Leftrightarrow Z_n=G$ for some finite n (same n)

in this case, we have $G^i<Z_{n-i}$ (G^i may be strict), $\forall 0\le i\le n$.

Pf first "$\Leftrightarrow$": same proof as for solvable groups.

second "$\Leftrightarrow$": want: $\begin{cases} G^n=\{1\}\Leftrightarrow Z_n=G \\ G^i<Z_{n-i} \end{cases}$

induction on n:

Consider $\pi: G \to G/Z(G) = \overline{G}$.

Claim $G^n = \{1\} \overset{①}{\Leftrightarrow} (\overline{G})^{n-1} = \{1\} \overset{\text{inductive hypothesis}}{\Longleftrightarrow} Z_{n-1}(\overline{G}) = \overline{G} \overset{②}{\Leftrightarrow} Z_n = G$.

for ①. $(\overline{G})^{n-1} \Leftrightarrow G^{n-1} < Z(G) \Leftrightarrow [G, G^{n-1}] = \{1\} \Leftrightarrow G^n = \{1\}$.

for ②. $Z_i = \pi^{-1}(Z_{i-1}(\overline{G}))$ essentially by definition. take $i = n$.

Also by inductive hypothesis $(\overline{G})^i < Z_{n-1-i}(\overline{G})$.

Taking π^{-1} gives $G^i < \pi^{-1}(\overline{G}^i) < \pi^{-1}(Z_{n-1-i}(\overline{G})) = Z_{n-i}$.

($G^i < \pi^{-1}(\overline{G}^i)$: may be strict; $\pi^{-1}(\overline{G}^i) < \pi^{-1}(Z_{n-1-i}(\overline{G}))$: inductive hypothesis)

REM The smallest $n \in \mathbb{N}$ for which $G^n = \{1\}$ is called nilpotency class (冪零类) of G

eg. $N = \left\{\begin{pmatrix} 1 & * & * \\ & 1 & * \\ & & 1 \end{pmatrix}\right\}$ is nilpotent of nilpotency class 2.

$G = \left\{\begin{pmatrix} * & * & * \\ & * & * \\ & & * \end{pmatrix}\right\}$ is solvable but not nilpotent. $G^0 = G$, $G^1 = [G,G] = N$, $G^2 = [G,N] = N$.

p-groups are nilpotent.

PROP Let P be a nontrivial p-group

1). $Z(P) \neq \{1\}$ (seen earlier via the class equation)

in particular, all p-groups are nilpotent.

2). If $H \lhd P$, $H \neq \{1\}$, then $H \cap Z(P) \neq \{1\}$.

3). If $H \lneq P$, then $H \lneq N_P(H)$

in particular, if $H < P$ has index p then $H \lhd P$ (since $N_P(H) = P$).

Pf 2) Consider $P \overset{\text{Ad}}{\curvearrowright} H$. $\#H = \sum \#(P/\text{Stab}_P(h_i))$ (div by p)

$\text{Stab}_P(h_i) = P \Leftrightarrow \forall x \in P,\ x h_i x^{-1} = h_i \Leftrightarrow h_i \in H \cap Z(P) \Rightarrow H \cap Z(P) \neq \{1\}$.

3). induction on $\#P$.

if $Z(P) \not\leq H$ then $Z(P) < N_P(H) \Rightarrow H \lneq N_P(H)$

if $Z(P) < H$ consider $\overline{H} = H/Z(P) < \overline{P} = P/Z(P)$.

inductive hypothesis $\overline{H} \subsetneq N_{\overline{P}}(\overline{H}) \Rightarrow H \subsetneq N_P(H)$. $\square$

THM (Characterization of finite nilpotent groups)

Let G be a finite group with $\#G = p_1^{r_1} \cdots p_m^{r_m}$ $P_i \in Syl_{p_i}(G)$

The followings are equivalent:

1) G is nilpotent

2) if $H \subsetneq G$ then $H \subsetneq N_G(H)$

3) all P_i are normal subgps of G

4) $G \simeq P_1 \times \cdots \times P_m$.

Pf 3) $\Rightarrow$ 4) $\checkmark$.

4) $\Rightarrow$ 1) since each P_i is nilpotent.

2) $\Rightarrow$ 3) Recall $N_G(N_G(P_i)) = N_G(P_i)$.

2) $\Rightarrow N_G(P_i) = G \Rightarrow P_i \lhd G$

1) $\Rightarrow$ 2) same proof as in PROP (G nilpotent $\Rightarrow G/Z(G)$ nilpotent). $\square$